

Name _____ Date _____ Period _____

Comparing All Seven Parent Functions

Algebra II | Unit 1 | Day 8 | TEK A2.2.A

Objective: I can compare the key attributes of all seven parent functions, name a function from its attributes alone, and find a relative maximum or minimum on a given interval.

Part 1 — The Master Comparison Table

Fill this in from your Day 4-7 notes. This is your reference sheet for the rest of the year.

Function	Domain	Range	Symmetry	Asymptotes
$\sqrt{x}$	_____	_____	_____	_____
$ x $	_____	_____	_____	_____
x^3	_____	_____	_____	_____
$\sqrt[3]{x}$	_____	_____	_____	_____
$1/x$	_____	_____	_____	_____
b^x	_____	_____	_____	_____
log base b	_____	_____	_____	_____

Part 2 — Intercepts and Extremes

Function	Intercept(s)	Minimum?	Maximum?
$\sqrt{x}$	_____	_____	_____
$ x $	_____	_____	_____
x^3 and $\sqrt[3]{x}$	_____	_____	_____
$1/x$	_____	_____	_____
b^x	_____	_____	_____
log base b	_____	_____	_____

Only _____ of the seven have a minimum, and _____ has a maximum. Only _____ has no intercept at all.

Part 3 — Name That Function

Use only the attributes listed. Name the function and sketch its shape in the box.

1. Domain and range both all reals except 0; two branches; asymptotes at $x = 0$ and $y = 0$.

Function: _____

2. Domain all reals, range $[0, \infty)$, folds across the y-axis, minimum of 0.

Function: _____

3. Domain $(0, \infty)$, range all reals, passes through $(1, 0)$.

Function: _____

4. Domain and range both $[0, \infty)$, no symmetry, no asymptotes.

Function: _____

Part 4 — Relative Max and Min on an Interval

A graph rises and falls between $y = 1.5$ and $y = -1.5$ over the interval from 0° to 360° .

Relative maximum on that interval: _____ Relative minimum: _____

Now consider $f(x) = x^3$ on the interval $[-2, 2]$.

Relative maximum: _____ Relative minimum: _____

x^3 has no overall max or min, but on a _____ it has both. Why? _____

Part 5 — Sort by One Attribute

Which parent functions have a restricted domain? _____

Which have any asymptote? _____

Which have range all real numbers? _____

Practice

Use your comparison tables. Show reasoning where there is room.

1. Two parent functions pass through $(1, 1)$ and $(0, 0)$. Name them. _____

2. Which parent function never touches either axis? _____

3. Which two are reflections of each other across $y = x$? Give two different pairs.

4. A graph has a minimum but no maximum, and no asymptotes. Name both possibilities.

5. $f(x) = |x|$ on the interval $[-3, 1]$: minimum _____ maximum _____

6. Explain why a function can have a relative maximum on an interval but none overall.

7. Which parent function has two separate branches? _____

8. True or false: every parent function with an asymptote also has a restricted domain.

9. Which two parent functions have both a minimum and pass through the origin? _____

Exit Ticket

Two parent functions have domain and range that are both all real numbers, and both have 180° rotational symmetry.

Name them: _____ Name one function with a restricted range:
